Coaching Session 2 Recall

Graphing

1. Sketch a region defined by an inequality in x and y. Decide which side of the boundary to shade and whether the boundary is included.

2. Sketch shifted parabolas that open vertically or horizontally. Complete the square when needed, and identify the vertex and direction of opening.

3. Sketch an inverse function by interchanging the roles of the x- and y-coordinates. Explain why switching the axes corresponds to reflecting the graph across $y = x$, and when a domain restriction is needed for the inverse to be a function.

4. Recognize cylinders, paraboloids, hyperboloids, ellipsoids, spheres, and cones from equations. Explain what an absent coordinate tells you about a surface.

5. Recognize rotational symmetry in a function that depends on x^2+y^2. Use its dependence on distance from the origin to distinguish a central peak, a circular ridge, a bowl, and radial oscillations.

Gradients

6. Compute the gradient of a function of two or three variables.

7. Find a directional derivative from a vector or angle, normalizing the supplied vector when necessary. Interpret its sign, magnitude, and units. Explain what happens when the direction is reversed.

8. Find the directions and rates of steepest ascent and descent, and all directions in which the instantaneous rate of change is zero. Include the case where the gradient is zero.

9. Use a contour plot to identify possible local maxima, local minima, saddle points, and points where the gradient is zero. Explain what nearby contour values tell you.

Vector Fields

10. Given a picture of vectors $\mathbf{u}$ and $\mathbf{v}$, draw combinations such as $\mathbf{v} - 2\mathbf{u}$. Show vector addition, subtraction, and scalar multiplication geometrically.

11. Sketch a vector field or match a formula to its plot. Use values at sample points, signs of components, zero vectors, and locations where arrows are horizontal or vertical.

12. Describe how multiplying a vector field by a constant scalar changes the directions and lengths of its arrows. Include positive, negative, and zero scalars.

13. Estimate gradients and sketch a gradient field from a contour plot. Explain the relationship to level curves, and compare gradient magnitudes and directional derivatives using contour values, spacing, and orientation.

14. Draw the unit circle with an orientation that keeps its interior on your left as you travel around it. Parameterize the circle with this orientation, and describe how to reverse it.

15. Predict the sign of a vector line integral from a field diagram and an oriented curve. Recognize when the sign is hard to determine from the diagram alone. Use the field's tangential component, possible cancellation, and the effect of reversing orientation.

Bounds of Integration

16. Sketch a region from integration bounds. Write bounds in both orders for a region described by curves, inequalities, or a diagram, splitting the region when necessary.

17. Set up and evaluate double integrals over nonrectangular regions. Reverse the order of integration when it makes evaluation easier.

18. Describe a region in polar coordinates and convert a Cartesian double integral to polar coordinates, including the area element. Choose Cartesian or polar coordinates based on the region and integrand.

19. Use polar coordinates to evaluate a double integral or a volume involving spheres, cylinders, or paraboloids. Determine radial and angular bounds from the geometry.